

## Solution Architecture

**Date:** 28 June 2026 **Team ID:** \_\_\_\_\_ **Project Name:** Rising Waters – AI Flood Prediction System

Maximum Marks: 5 Marks

### Solution Architecture Diagram

Provide a high-level visual representation of your solution architecture. Fill in the technologies and components in the boxes below.

|                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------|
| Presentation / Client Layer<br>HTML<br>CSS<br>JavaScript<br>Bootstrap                                                               |
| ↓ ↑                                                                                                                                 |
| API Gateway / Load Balancer<br>Flask Web Server                                                                                     |
| ↓ ↑                                                                                                                                 |
| Application Layer:<br>Flask Application<br>Python<br>Request Handling<br>Prediction Controller                                      |
| ↓ ↑                                                                                                                                 |
| Auth Service<br>Not Implemented                                                                                                     |
| ↓ ↑                                                                                                                                 |
| Core Logic Service<br>XGBoost Machine Learning Model<br>Flood Prediction Engine<br>Risk Analysis<br>Safety Recommendation Generator |
| ↓ ↑                                                                                                                                 |
| External API Integrations<br>Currently Not Used<br>(Future: Weather API / Rainfall API)                                             |
| ↓ ↑                                                                                                                                 |
| Data / Storage Layer<br>SQLite Database<br>Prediction History<br>CSV Dataset<br>Pickle Model Files (.pkl)                           |

↓ ↑

### Component Description Table

| Component Name     | Description / Role in Architecture                                      | Technologies Used                |
|--------------------|-------------------------------------------------------------------------|----------------------------------|
| Presentation Layer | User interface for entering flood parameters and displaying predictions | HTML, CSS, JavaScript, Bootstrap |
| API Gateway        | Handles HTTP requests between client and server                         | Flask                            |
| Microservices      | Executes prediction logic and processes user inputs                     | Python, XGBoost                  |
| Database           | Stores prediction history and model-related data                        | SQLite, CSV Dataset              |